

Hidden Dimensions of Quality of Life for the Elderly

by Dr. Josip Derado, Shelly Ellis and Muniza Naqvi

Abstract

According to numbers reported by the Administration for Community Living at the U.S. Department of Health Services, the Elderly (65 yrs. and older) represent 14.5% of the total U.S. population¹, which is about one in every seven Americans. This number is expected to rise with the aging baby boomers and ongoing advances in medical care. Understanding aging trends and the functioning of adults later in life may help address their needs more effectively, reduce “aging” costs, and improve their care. A study, called the National Health and Aging Trends (www.NHATS.org) was conducted by John Hopkins University with support from the National Institute of Aging, to better understand issues pertaining to mobility of the individuals and their economic conditions, as well as information on their physical and mental health. We chose Round 5, where there were 45 Data Collection instrument groups provided. The goal of our analysis was to understand what constructs underlie the data using Factor Analysis by reducing the given 1,495 variables into describable lesser dimensions. Our analysis indicated that the three most important measures of quality of life are mobility, financial status, and level of independence.

Introduction

NHATS gathered information on a nationally representative sample of Medicare beneficiaries ages 65 and older. The sample was refreshed periodically so that researchers could study national-level disability trends as well as individual trajectories. Annual, in-person interviews we conducted to collect detailed information on the disablement process and its consequences. The original data set contained 1492 variables and 8334 observations. The variables were separated into 45 subgroups. Exploring all 45 subgroups would have proved highly cumbersome. Instead a few groups were selected based on the general questions that they addressed. These included the following subgroups: Quality of Care in Last Month(LM), Rehabilitation(RH), Economic Well-being(EW), Household Activities(HA), Income and Assets(IA), Health Conditions(HC), Self-Care Activities(SE), Service Environment(SE). Income and Assets primarily considered how well the participant managed their personal finances. Economic Well-being also considered how the participant was doing financially, for example, questions regarding total credit card debt or did the participant need help from a family member; however, the questions focused on external resources of assistance. Household Activities considered activities done in the home, for example was how laundry got done and did the participant need assistance, etc. Self-Care Activities focused mainly on how well the participants were able to take care of themselves personally.

Methodology

Data Clean Up

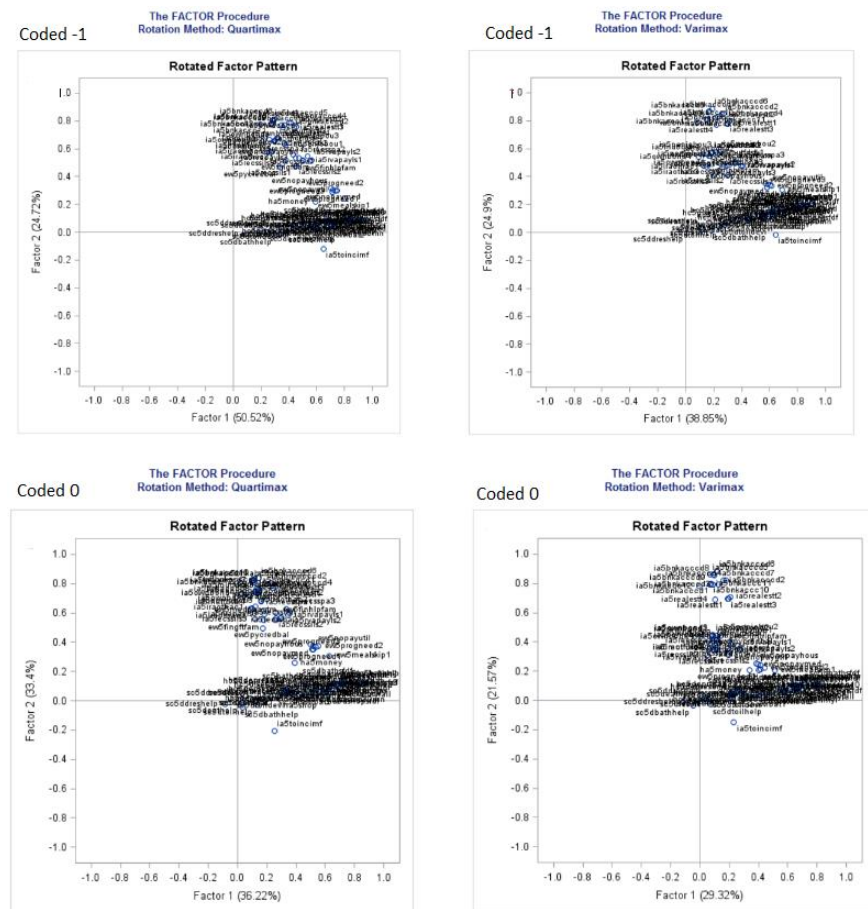
Variables with more than 2000 observations missing (about 25% missing), or answered “Inapplicable” or “DK” for “Don’t Know”) along with other values which were coded negative were eliminated. This left us with 117 variables and 6732 observations.

Furthermore, different coding formats were tested to see their impact on factor separation. We changed the coding for the value of ‘Inapplicable’ from ‘-1’ to ‘0’, and came to the conclusion that the response ‘Inapplicable’ coded as ‘0’ yields better separation between the factors and significantly decreases the number of cross-loading variables. Figure 1 shows the separation between the Factor 1 and Factor 2 using both versions of coding for Varimax and Quartimax rotations. We did not

¹ "Aging Statistics." N.p., 24 May 2016. Web. 22 Apr. 2017. <https://aoa.acl.gov/Aging_Statistics/Index.aspx>.

investigate further coding changes, but it seems that the factor analysis is quite sensitive to how data is coded. Coding balanced in its responses seems to have the best results.

Figure 1: Coding Impact on Factor Patterns



Modeling Assumptions

For this analysis we wanted to conduct factor analysis and also compare it to results from principal component analysis. Both methods require the following assumptions:

- Variables that can be measured at a continuous level, which our data meets since it has ordinal variables with most variables following a Likert Scale.
- Variables have a linear relationship – this is assumed.
- The sample size must be adequate as their methods depend on correlation matrices, and correlations take large sample sizes to stabilize. This criterion is easily met with 6732 observations, but is also verified with Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy (MSA) of 0.9564.
- Data should be suitable for reduction meaning it should have enough variables, which are moderately correlated to allow for dimension reduction. Our data originally had 1,495 variables. Even when looking at only a few subgroups we still have 117 variables, so this criterion is also met.
- There should be no significant outliers – this is assumed.

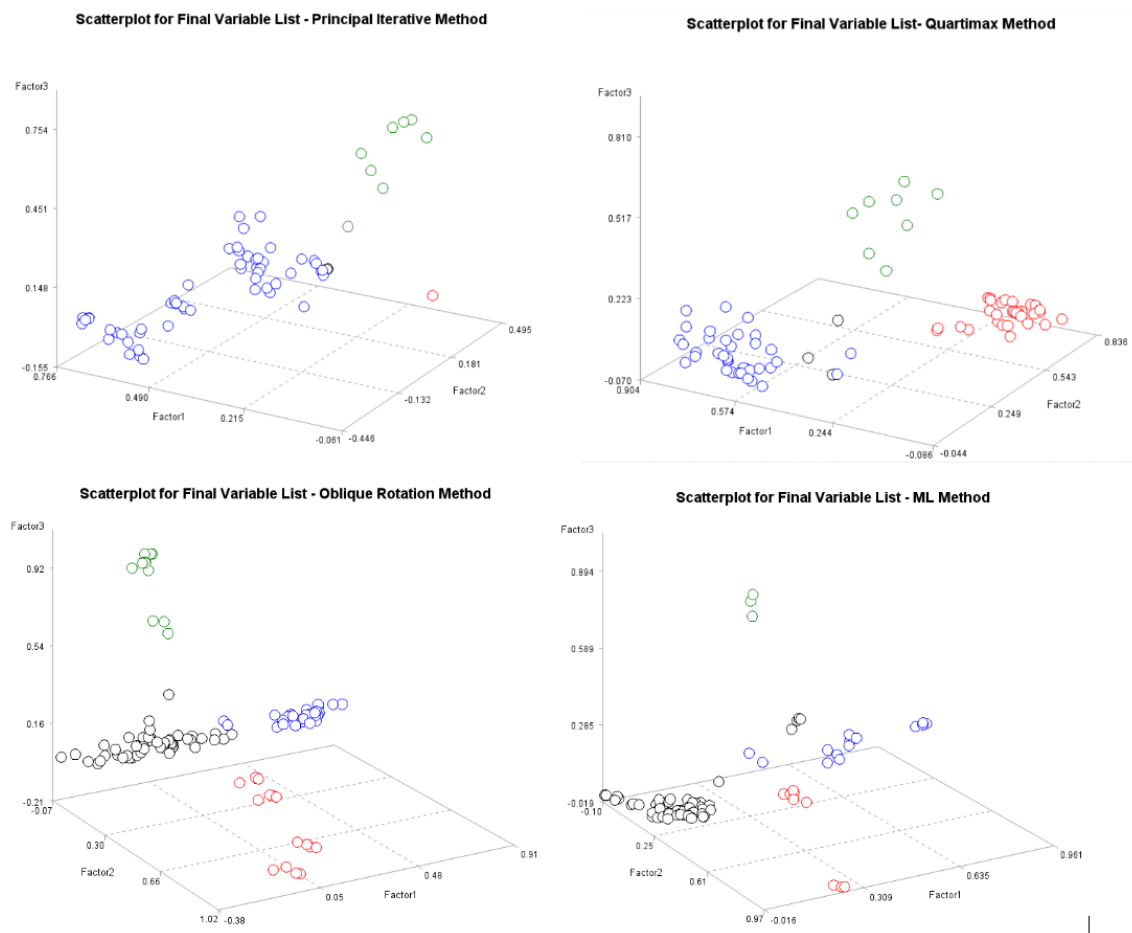
Factor Analysis also requires data from a homogenous population, which is satisfied because our sample is selected from 65+ year olds who are on Medicaid. We also assumed that:

- Specific factors or random errors, and common factors have mean zero.
- Common factors have variance one.
- Common factors and specific factors are uncorrelated among themselves and with one another.

Choosing a Factor Extraction Method

We ran factor analysis with a variety of rotational methods, which included both orthogonal (Varimax and Quartimax), and oblique (Harris-Kaiser) rotations. We also compared this to principal component analysis, which suggested 30 components. We opted to use factor analysis with principal iterative method and the Quartimax rotation method. This was decided based on the quality of the separation between factors, the number of factors, and the number of cross-loading variables (the variables which contain cross-loadings). Factor 1 vs. Factor 2 Pattern Plots can be seen in Appendix I: Figure 8 through Appendix I: Figure 12. Although HK has the best separation, it also retains 10 factors. Since the philosophy behind this analysis is to find the best parsimonious model, Quartimax with three factors was instead chosen. In general, all methods yield mathematically significant results, but the real purpose of conducting Factor Analysis is to find a describable relationship among variables. For a comparison between the different methods see Figure 2, which shows scatterplots of the first three factors for each method where the blue dots represent variables belonging to the Factor1, red to the Factor 2, green to the Factor3 and black dots represent variables belonging to all the other factors. The principal component method returns one essential factor, thus offering no meaningful factor analysis. On the other hand, ML method (while assuming multivariate normality) and the oblique rotation returned too many factors with few variables in each factor, which again did not offer any meaningful interpretation. In addition, the oblique rotation by the nature of the method is harder to interpret. Thus the orthogonal rotations method Varimax and the Quartimax were given the best factor analysis. The results were similar except the fact that the Quartimax returned less cross-loading variables. The Quartimax orthogonal rotation matrix is given in Appendix I: Figure 3.

Figure 2: Scatterplot

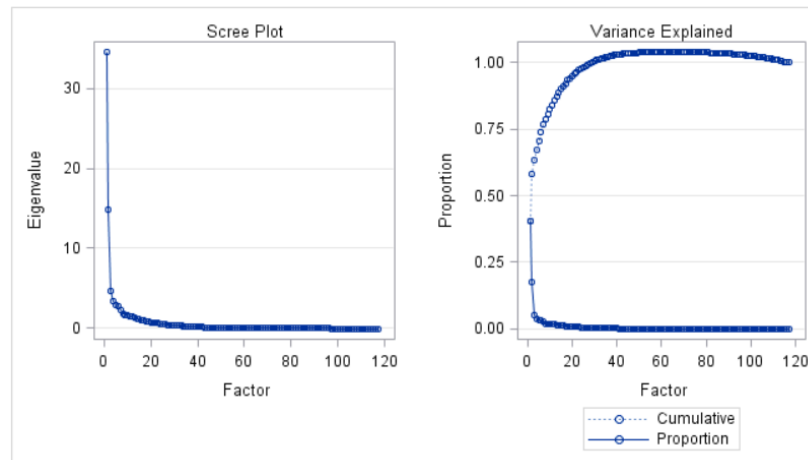


Number of Factors Retained

The initial number of factors was decided upon using the factor principal component analysis. The Scree plot in Figure 3 suggested between 8 and 12 factors. The number of the eigenvalues higher than 1 suggested 16 factors and the cumulative percentage of the explained variance with cutoff at 80% asked for 9 factors (See Appendix I: Figure 1). The Chi-Square

Significance Test suggested that 10 factors are sufficient (see Appendix I: Figure 2). In order to achieve a parsimonious model 10 factors for each method were tested.

Figure 3: Scree Plot based on Factor Principal Component Method



The final number of factors for variable partition were decided using the following rules:

The variables with one dominant loading of absolute value greater than 0.50, and no other loadings larger in absolute value than 0.30 were considered to belong to the factors of their highest loading. The variables with two or more loadings of absolute value higher than 0.3 were considered cross-loading variables. The cross-loading variables which had loadings higher than 0.50, and for which the difference between the two highest loadings was significant (>0.2) were considered to belong to the factor of the highest loading as well. The rest of the cross-loading variables, as well as all the other variables with highest absolute loading smaller than 0.50 were eliminated from the analysis. Following this process, all the variables from the subgroups LM (Quality of Care in Last Month) and SE (Service Environment) were eliminated.

Finally, the factors with two or less variables were also discarded from the final analysis as we they are generally considered weak and unstable. In the end we ended up with 3 factors 75 variables, while we eliminated 28 variables, and three factors which contained less than three variables.

Results

Complete results for all factors retained can be seen in Appendix I: Figure 6. Next, the most important variables for each factor are discussed.

Factor 1: The Mobility Factor

Figure 4 shows the top ten loadings for Factor 1. The different colors show group separation. In total, Factor 1 combines 33 variables from four subgroups; Health Conditions, Household Activities, Self-Care Activities and Rehabilitation (not shown in top ten loadings). The variable discussing broken or fractured bones has the highest loading for Factor 1. This considered with the other variables listed here, such as whether they needed meals delivered indicates that the level of mobility maintained by the elderly has a significant impact on their quality of life. We also see some health variables in Factor 1 since mobility is directly affected by health conditions.

Figure 4: Factor 1 Top 10 Loadings

Subgroup	Variable Name	Maxload	Question (Reduced Form)	Highest Frequency Answers
Health Conditions	hc5brokebon1	90.4%	Have a broken or fractured hip, or other broken or fractured bones?	No - 88.19%
Household Activities	ha5mealwheel	87.8%	In the last month, when you had hot meals, how often were they from Meals on Wheels or other food assistance programs?	Never - 85.06%
Self-Care Activities	sc5deatsfdf	87.0%	In the last month, when you used adapted utensils, how much difficulty did you have eating by yourself and without help?	No Difficulty by self - 85.04%
Self-Care Activities	sc5dresyrgo	86.6%	Compared to a year ago, do you now get dressed more often, less often, or about the same?	Same - 84.7%
Self-Care Activities	sc5dtoilsfdf	86.3%	In the last month, how much difficulty did you have using the toilet by yourself and without help?	No Difficulty by self - 81.79%
Self-Care Activities	sc5bathyrgo	83.8%	Compared to a year ago, do you now shower/take baths/ wash up more often, less often, or about the same?	Same - 81.65%
Health Conditions	hc5flsinmth	81.7%	In the last month, have you fallen down?	No - 80.23%
Self-Care Activities	sc5dbathsfdf	81.1%	In the last month, when you used your grab bars/shower or tub seat how much difficulty did you have showering/taking a bath without help?	No Difficulty by self - 74.62%
Self-Care Activities	sc5bathhlp	79.0%	In the last month, did anyone ever help you shower/ take a bath/wash up?	No - 81.1%
Health Conditions	hc5disescn10	74.9%	Have you been diagnosed with cancer?	No - 76.63%

Factor 2: The Money Factor

Figure 5 shows the top ten loadings for Factor 2. Factor 2 only captures a total of 30 variables, all from the subgroup Income and Assets. Factor 2 indicates that majority of people surveyed don't have savings, CD's, real estate or any other substantial financial assets. On the other hand, they are not in debt either as 45.7% have paid off their credit card debts, and 22% don't even have credit cards. It seems like these people have either cashed-in their assets, and are enjoying the fruits of their labor. Or that they either didn't have very much to begin with or lost substantial financial gains.

Figure 5: Factor 2 Top 10 Loadings

Subgroup	Variable Name	Maxload	Question (Reduced Form)	Highest Frequency Answers
Income and Assets	ia5bnkacccd6	83.6%	Do you or your spouse own a savings account?	No - 78.27%
	ia5bnkacccc10	82.5%	Do you or your spouse own any certificate of deposits?	No - 81.98%
	ia5bnkacccc11	82.0%	Does your spouse own any certificate of deposits?	No - 78.37%
	ia5bnkacccd7	82.0%	Does your spouse own a savings account?	No - 62.97%
	ia5bnkacccd9	81.9%	Do you own any certificate of deposits?	No - 75.69%
	ia5bnkacccd5	81.7%	Do you own a savings account?	No - 52.03%
	ia5bnkacccd8	81.0%	Do you own a savings account?	No - 55.53%
	ia5bnkacccc12	80.6%	You don't own a any certificate of deposits, correct?	Yes - 69.45%
	ia5bnkacccd4	80.2%	Do you own a checking account?	No - 80.48%
	ia5bnkacccd2	80.0%	Does your spouse own a checking account?	No - 76.29%

Factor 3: The Independence Factor

Figure 6 shows the top ten loadings for Factor 3. Factor 3 indicates that majority of the surveyed people did not require assistance in their last month, and maintained their independence. They were all able to mostly maintain their autonomy, except for one factor – majority of the surveyed people required toileting devices.

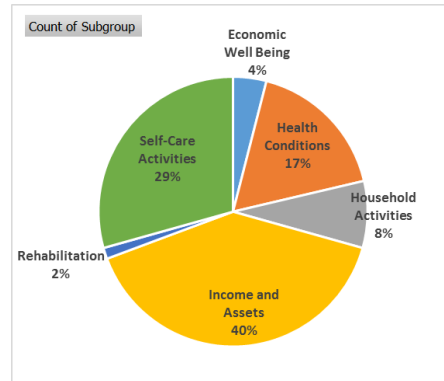
Figure 6: All Loadings for Factor 3

Subgroup	Variable Name	Maxload	Question (Reduced Form)	Highest Frequency Answers
Self-Care Activities	sc5deathelp	81.0%	Did anyone help you eat?	No help eating in last month - 86.62%
	sc5ddreshelp	79.4%	Did anyone help you get dressed?	No help dressing in last month - 78.43%
	sc5dbathhelp	75.6%	Did anyone help you get bathe or shower?	No help bathing in last month - 81.05%
	sc5dtoilhelp	72.4%	Did anyone help you use the toilet?	No help toileting in last month - 88%
	sc5ddresdevi	64.8%	Did you use any dressing devices to help get dressed?	No use of dressing devices in last month - 88.98
	sc5deatdevi	64.3%	Did you use any adapted utensils to help you eat?	No use of adapted utensils in last month - 93.34%
	sc5dtoildevi	52.0%	Did you use any toileting devices to use the toilet?	Use of toileting devices in last month - 49.3%

Conclusion

We were seeking a simple structure within provided data, which would reduce the data's dimensionality by identifying latent variables reflected in observed variables. Our analysis indicated that the three most important measures of quality of life are mobility, financial status and level of independence of the respondents. These are the identified latent variables. Our final model included 75 total variables, 24 of which were cross-loadings and were included in the maximum factor. Appendix I: Figure 6 shows the complete list of variables retained (items in grey are cross-loadings). All of the variables have one thing in common. Each question relates to the individual's ability to do the basic things that are necessary to live. For example, getting dressed, eating, bathing and being able to move around without assistance. To investigate the impact of each subgroup (observable variables) in our analysis we created the Pie Chart in Figure 7. It is evident that Income and Assets have the highest impact, followed by Self-care activities and Health Conditions on the quality of life for the 65 and older population.

Figure 7: Subgroup Contribution to Quality of Life



Contributions

Dr. Josip Derado – SAS Programming, analysis and interpretation, methodology.

Shelly Ellis – Helped with analysis, interpreting results and writing the paper.

Muniza Naqvi – Analysis, model reduction and comparison. Helped interpret results, organize and write paper.

Appendix I – Output

Appendix I: Figure 1 – Eigenvalues for Factor Principal Components Method (no rotation)

Preliminary Eigenvalues: Total = 85.273683 Average = 0.7288349				
	Eigenvalue	Difference	Proportion	Cumulative
1	34.5899461	19.7015924	0.4056	0.4056
2	14.8883536	10.2831841	0.1746	0.5802
3	4.6051695	1.2545792	0.0540	0.6342
4	3.3505904	0.4799294	0.0393	0.6735
5	2.8706610	0.0725394	0.0337	0.7072
6	2.7981215	0.5907406	0.0328	0.7400
7	2.2073810	0.4663949	0.0259	0.7659
8	1.7409860	0.1439700	0.0204	0.7863
9	1.5970161	0.0204657	0.0187	0.8050
10	1.5765503	0.0526061	0.0185	0.8235
11	1.5239442	0.0116402	0.0179	0.8414
12	1.5123040	0.2120913	0.0177	0.8591
13	1.3002127	0.1884684	0.0152	0.8744
14	1.1117444	0.0343177	0.0130	0.8874
15	1.0774266	0.0594871	0.0126	0.9000
16	1.0179396	0.1000760	0.0119	0.9120
17	0.9178635	0.0390301	0.0108	0.9227

Appendix I: Figure 2 - Chi-Square Significance Test for Factor Principal Components Method

Significance Tests Based on 6732 Observations			
Test	DF	Chi-Square	Pr > ChiSq
H0: No common factors	6786	1029498.15	<.0001
HA: At least one common factor			
H0: 10 Factors are sufficient	5661	264771.719	<.0001
HA: More factors are needed			

Appendix I: Figure 3 - Quartimax Model Rotation Matrix

The FACTOR Procedure
Rotation Method: Quartimax

Orthogonal Transformation Matrix										
	1	2	3	4	5	6	7	8	9	10
1	0.73572	0.65409	0.03570	0.08790	0.11328	0.00820	0.06026	0.05180	0.03922	0.03361
2	0.64669	-0.74507	0.10004	-0.05058	0.01665	-0.02777	0.07239	0.05570	0.04822	0.04873
3	-0.10056	0.08040	0.93587	-0.18817	-0.13842	0.01532	0.08052	0.13538	0.08735	0.14259
4	-0.08461	-0.10038	0.25746	0.76033	0.55643	0.08562	-0.13427	-0.04449	-0.00480	-0.03756
5	-0.08612	-0.00397	-0.08084	-0.11513	0.35697	-0.07114	0.46720	0.38702	-0.56596	0.38977
6	0.01946	-0.02118	-0.05948	0.10233	-0.23124	0.94039	0.02951	0.18758	-0.07408	0.07619
7	-0.00458	-0.00373	-0.05150	0.57887	-0.64024	-0.30499	0.20175	0.33061	-0.01727	0.09479
8	-0.04019	-0.00271	0.04233	0.06143	0.01726	0.09047	0.81598	-0.35590	0.12908	-0.41854
9	-0.03021	0.00196	-0.08009	0.08395	-0.03748	0.03115	0.13714	-0.49995	0.28960	0.79411
10	-0.11354	-0.00177	-0.16017	-0.08105	0.25613	0.00470	0.14877	0.55047	0.74957	0.04752

Appendix I: Figure 4 - Brief Extract of Quartimax Eigenvalues

Eigenvalues of the Reduced Correlation Matrix: Total = 68.7958486 Average = 0.58799871				
	Eigenvalue	Difference	Proportion	Cumulative
1	34.4461794	19.7181583	0.5007	0.5007
2	14.7280211	10.3522269	0.2141	0.7148
3	4.3757941	1.2114922	0.0636	0.7784
4	3.1643020	0.4006286	0.0460	0.8244
5	2.7636734	0.1848273	0.0402	0.8646
6	2.5788461	0.5169231	0.0375	0.9020
7	2.0619230	0.3056502	0.0300	0.9320
8	1.7562728	0.2301613	0.0255	0.9575
9	1.5261114	0.1310271	0.0222	0.9797
10	1.3950843	0.0867786	0.0203	1.0000
11	1.3083058	0.0948925	0.0190	1.0190
12	1.2134132	0.1814515	0.0176	1.0367
13	1.0319617	0.1347858	0.0150	1.0517
14	0.8971759	0.0469870	0.0130	1.0647
15	0.8501889	0.0841432	0.0124	1.0771

Appendix I: Figure 5 - Variance explained by each Factor using Quartimax

Variance Explained by Each Factor									
Factor1	Factor2	Factor3	Factor4	Factor5	Factor6	Factor7	Factor8	Factor9	Factor10
24.915245	22.974640	4.314934	3.069136	2.939103	2.540143	2.206051	1.981295	1.961802	1.893858

Appendix I: Figure 6 - Complete List of Variables per Factor. Variables in grey are crossloadings

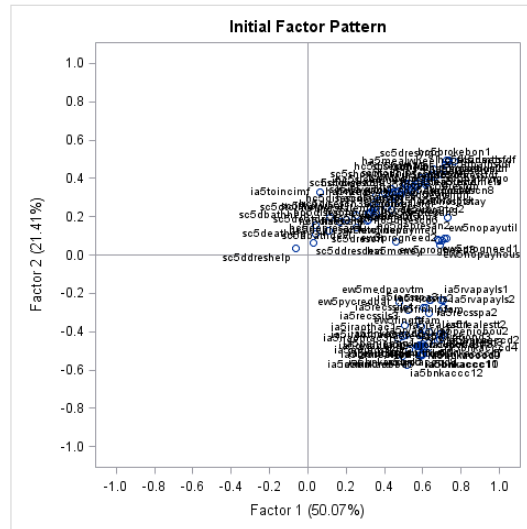
	Subgroup	Variable Label	Variable Name	Maxload
Factor 1	Health Conditions	R5 HC5A SP BROKEN OR FRACT HIP	hc5brokebon1	90.4%
		R5 HC14 FALL DOWN IN LAST MONTH	hc5fillslmth	81.7%
		R5 HC2 SP HAS CANCER	hc5disescn10	74.9%
		R5 HC5B SP OTHR BRKN FRACT BONE	hc5brokebon2	72.6%
		R5 HC2 SP HAD STROKE	hc5disescn8	72.1%
		R5 HC2 SP HAD HEART ATTACK	hc5disescn1	71.6%
		R5 HC15 WORRIED ABOT FALLNG DOWN	hc5worryfall	70.3%
		R5 HC7 SP HOSP STAY LAST 12MOS	hc5hosptstay	66.0%
		R5 HC22 OFTN MEDICATE HELP SLEEP	hc5sleepmed	65.2%
		R5 HC21 TROBLE FALLNG BCK ASLEEP	hc5trbfbck	62.4%
		R5 HC20 OVER 30 MIN FALL ASLEEP	hc5aslep30mn	59.1%
		R5 HC17 FALLEN DOWN IN 12 MONTHS	hc5faleninyr	55.6%
		R5 HC2 SP HAS DEMENTIA OR ALZH	hc5disescn9	52.0%
	Household Activities	R5 HA30C OFTN GET MEALS ON WHEEL	ha5mealwheel	87.8%
		R5 HA30B OFTN HAV RESTARNT MEALS	ha5restamels	71.1%
		R5 D LAUNDRY SELF AND DIFF LEVEL	ha5dlaunsfdf	65.6%
		R5 D SELF BANKS AND DIFF LEVEL	ha5dbanksfdf	63.3%
		R5 D SLF SHOPPING AND DIFF LEVEL	ha5dshopsfdf	59.6%
	Rehabilitation	R5 D SELF MEALS AND DIFF LEVEL	ha5dmealsfdf	58.5%
		R5 RH1 RECEIVED REHAB IN LAST YEAR	rh5rehab	72.7%
	Self-Care Activities	R5 D DIFF EATING BY SELF WO HELP	sc5deatsfdf	87.0%
		R5 SC26 HOW OFTN YOU GET DRESSED	sc5dresyrgo	86.6%
		R5 D DIFF LEVEL TOILETING SELF	sc5dtoilsfdf	86.3%
		R5 SC14 BATHE MRE OFT THN YR AGO	sc5bathyrgo	83.8%
		R5 D DIFF BATHING SELF NO HELP	sc5dbathsfdf	81.1%
		R5 SC11 ANYN HLP SHOWR BATH OTHR	sc5bathhlp	79.0%
		R5 D DIFF LEVEL DRESSING SELF	sc5ddressfdf	72.6%
		R5 SC3 DID ANYONE HELP YOU EAT	sc5eathlp	70.7%
		R5 SC23 ANYONE HELP GET DRESSED	sc5dreshlp	69.0%
		R5 SC17 ANYON HLP YOU USE TOILET	sc5toilhlp	67.3%
		R5 SC22 USE SPECL ITEMS TO DRESS	sc5dresdev	61.8%
		R5 SC16 SP USED PORTABLE COMMODE	sc5usvarto1	59.7%
		R5 SC5 DIFFICULTY EATING BY SELF	sc5eatslfdif	59.5%
		R5 SC16 SP USED PADS UNDERGMT	sc5usvarto2	55.5%
		R5 SC19 DIFFICULTY TOILETING SELF	sc5toildif	50.0%
Factor 2	Economic Well Being	R5 EW5 MED BILLS PAID OVERTIME	ew5medpaovtm	56.6%
		R5 EW7 FINANCIAL HELP FRM FAMILY	ew5finhlpfam	56.5%
		R5 EW11 FINANCIAL GIFTS TO FAMILY	ew5fingftfam	54.9%
	Income and Assets	R5 IA10 SPOUSE OWNS SAVING ACCT	ia5bnkacccd6	83.6%
		R5 IA10 SPOUSE OWNS CDS	ia5bnkacccc10	82.5%
		R5 IA10 SP SPOUSE OWN CDS	ia5bnkacccc11	82.0%
		R5 IA10 SP SPOUSE OWN SAVNG ACT	ia5bnkacccd7	82.0%
		R5 IA10 SP OWNS CDS	ia5bnkacccd9	81.9%
		R5 IA10 SP OWNS SAVINGS ACCT	ia5bnkacccd5	81.7%
		R5 IA10 NO SAVINGS ACCT OWNED	ia5bnkacccd8	81.0%
		R5 IA10 NO CDS OWNED	ia5bnkacccc12	80.6%
		R5 IA10 NO CHECK ACCT OWNED	ia5bnkacccd4	80.2%
		R5 IA10 SPOUSE OWNS CHECK ACCT	ia5bnkacccd2	80.0%
		R5 IA13 SPOUSE OWNS REAL ESTATE	ia5realestt2	78.1%
		R5 IA10 SP SPOUSE OWN CHECK ACCT	ia5bnkacccd3	77.8%
		R5 IA10 SP OWNS CHECK ACCT	ia5bnkacccd1	77.2%
		R5 IA13 SP OWNS REAL ESTATE	ia5realestt1	77.0%
		R5 IA13 SP SPOUSE OWN REAL ESTTE	ia5realestt3	76.9%
		R5 IA13 NO REAL ESTATE OWNED	ia5realestt4	75.6%
		R5 IA9 SPOUSE OWNS BONDS	ia5ownbond2	74.7%
		R5 IA9 SP SPOUSE OWN BONDS	ia5ownbond3	74.5%
		R5 IA8 SPOUSE OWNS FUNDS STOCK	ia5mutfdstk2	74.5%
		R5 IA9 SP OWNS BONDS	ia5ownbond1	74.2%
		R5 IA8 SP SPOUSE OWN FUNDS STOCK	ia5mutfdstk3	74.0%
		R5 IA8 SP OWNS MUTUAL FUND STOCK	ia5mutfdstk1	73.6%
		R5 IA9 NO BONDS OWNED	ia5ownbond4	73.2%
		R5 IA8 NO FUNDS OR STOCK OWNED	ia5mutfdstk4	71.7%
		R5 IA6 SPOUSE HAS PENSION PLAN	ia5penjobou2	69.8%
		R5 IA6 NO PENSION PLAN	ia5penjobou3	68.4%
		R5 IA6 SP HAS PENSION PLAN	ia5penjobou1	67.7%
		R5 IA7 SPOUSE HAS IRA OTHR ACC	ia5iraothac2	64.2%
		R5 IA7 SP HAS IRA OTH RETIRE ACC	ia5iraothac1	63.4%
		R5 IA7 NO IRA OTHR RETIRE ACCT	ia5iraothac3	62.5%
Factor 3	Self-Care Activities	R5 D HAS HELP EATING	sc5deathelp	81.0%
		R5 D HAS HELP WHILE DRESSING	sc5dreshelp	79.4%
		R5 D HAS HELP WHILE BATHING	sc5dbathhelp	75.6%
		R5 D HAS HELP WHILE TOILETING	sc5dtoilhelp	72.4%
		R5 D USES DEVICES WHILE DRESSING	sc5ddresdevi	64.8%
		R5 D USES DEVICES WHILE EATING	sc5deatdevi	64.3%
		R5 D USES DEVICE WHILE TOILETING	sc5dtoildevi	52.0%

Appendix I: Figure 7 - Brief Extract of Quartimax Method Communalities

NAME_	LABEL_	Communalities
hc5brokebon1	R5 HC5A SP BROKEN OR FRACT HIP	0.849492073
ia5bnkacccd6	R5 IA10 SPOUSE OWNS SAVING ACCT	0.840376453
sc5deatsfdf	R5 D DIFF EATING BY SELF WO HELP	0.837738171
sc5dtoilsfdf	R5 D DIFF LEVEL TOILETING SELF	0.821262661
ew5mealskip1	R5 EW16 SKIP MEALS NO MONEY	0.816434837
ew5progneed2	R5 EW15 SP REC OTHR FOOD ASST	0.814206446
ia5bnkacccd4	R5 IA10 NO CHECK ACCT OWNED	0.810649237
ia5bnkacccd7	R5 IA10 SP SPOUSE OWN SAVNG ACT	0.805561455
ew5nopayutil	R5 EW18 NO MONEY FOR UTILITIES	0.799816935
ha5mealwheel	R5 HA30C OFTN GET MEALS ON WHEEL	0.798333024
ia5bnkacccd2	R5 IA10 SPOUSE OWNS CHECK ACCT	0.797699145
ia5bnkacccd5	R5 IA10 SP OWNS SAVINGS ACCT	0.793758822
ia5bnkacccd8	R5 IA10 NO SAVINGS ACCT OWNED	0.782095993
ew5nopaymed	R5 EW18 NO MONEY FOR MEDICAL	0.778204393
sc5dresyrgo	R5 SC26 HOW OFTN YOU GET DRESSED	0.775523532
ia5recsspa3	R5 IA1 NO SOC SECURTY PAYMNT REC	0.770729101
ia5mutfdstk2	R5 IA8 SPOUSE OWNS FUNDS STOCK	0.766460636
sc5bathhlp	R5 SC11 ANYN HLP SHOWR BATH OTHR	0.765096533
ia5bnkacccc10	R5 IA10 SPOUSE OWNS CDS	0.764402435
ia5ownbond2	R5 IA9 SPOUSE OWNS BONDS	0.760374902
ia5mutfdstk3	R5 IA8 SP SPOUSE OWN FUNDS STOCK	0.759367804
ia5bnkacccc11	R5 IA10 SP SPOUSE OWN CDS	0.755285018
ia5ownbond3	R5 IA9 SP SPOUSE OWN BONDS	0.75394428
ia5bnkacccd9	R5 IA10 SP OWNS CDS	0.752262862
ia5mutfdstk1	R5 IA8 SP OWNS MUTUAL FUND STOCK	0.751026709
sc5dbathsfdf	R5 D DIFF BATHING SELF NO HELP	0.750445735
ia5ownbond1	R5 IA9 SP OWNS BONDS	0.750204221

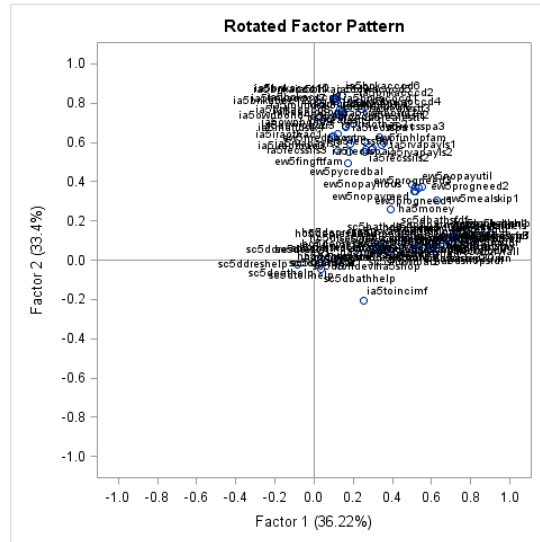
Appendix I: Figure 8 - Initial Factor Patterns (no Rotation)

The FACTOR Procedure
Initial Factor Method: Iterated Principal Factor Analysis



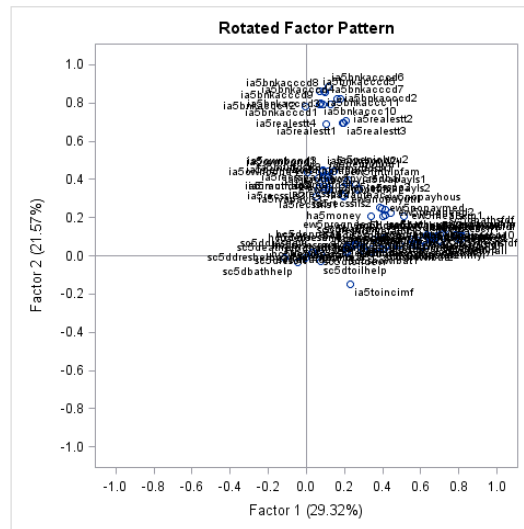
Appendix I: Figure 9 - Quartimax Rotation Factor Pattern

The FACTOR Procedure
Rotation Method: Quartimax



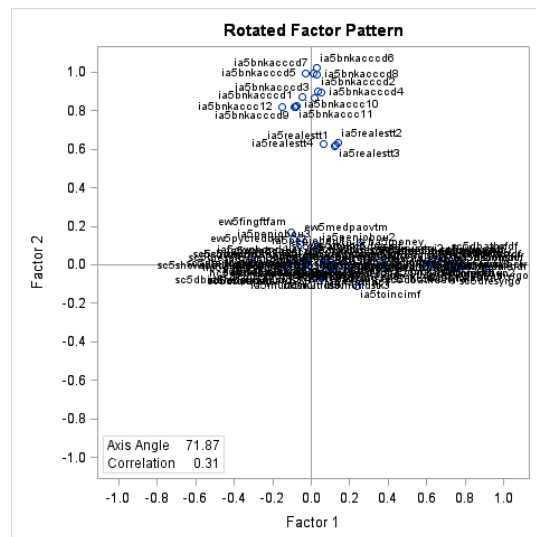
Appendix I: Figure 10 - Varimax Rotation Factor Pattern

The FACTOR Procedure
Rotation Method: Varimax

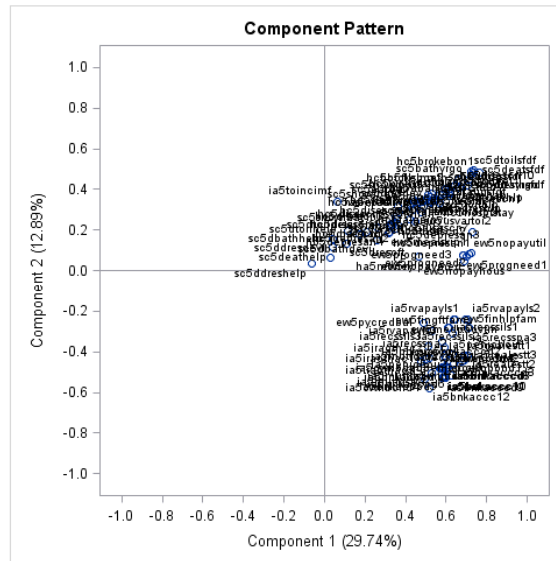


Appendix I: Figure 11 - HK Rotation Factor Pattern

The FACTOR Procedure
Rotation Method: Harris-Kaiser (hkpower = 0)



Appendix I: Figure 12: Principal Components 1 vs 2
The PRINCOMP Procedure



Appendix II – SAS Code

```
*SAS CODE FOR NHATS FACTOR ANALYSIS;
data trtba;
    set sasuser.nhats5;
run;
data focus;
    set trtba;
    keep lm: se: rh: ew: ha: ia: hc: sc;;
run;
proc contents data=focus; run;
* INITIAL CLEANING OF THE DATA- ELIMINATING VARIABLES;
*WITH HIGH NUMBER OF MISSING DATA AND OTHER VALUES OF NO INTEREST;
proc transpose data=trtba out=tranba;
run;
data tranp3;
    set tranba;
    array h{8334} col1-col8334;
    counter=0; max=0;
    do i = 1 to 8334;
        if h[i] < 0 then counter+1;
        if h[i]>max then max=h[i];
    end;
    if counter > 2000 then delete;
    if max>200 then delete;
    drop counter i
run;
proc transpose data=tranp3 out=trtp3;
run;
data focus;
    set trtp3;
    keep lm: se: rh: ew: ha: ia: hc: sc;;
run;
proc contents data=focus; run;
proc contents data=trtp3; run;
data trtp3;
    set trtp3;
    array h{*} r5dcontnew--ir5condhome6;
    do i = 1 to dim(h);
        if h[i]= -1 then h[i]=0;
        if h[i] = -9 then delete;
    end;
run;

*FACTOR ANALYSIS DATA ARE IN TRTP3;
*Testing Orthogonal Rotations;
proc factor data=trtp3 method=prininit nfactors=10 rotate=quartimax msa corr priors=smc hey outstat=quart02
plots=(scree initloadings preloadings loadings);
var lm: rh: ew: ha: ia: hc: sc: se;;
run;
```

```

proc factor data=trtp3 method=prininit nfactors=10 rotate=varimax msa corr priors=smc hey outstat=varimax02
plots=(scree initloadings preloadings loadings);
var lm: rh: ew: ha: ia: hc: sc: se;;
run;
*Testing Oblique Rotations;
proc factor data=trtp3 method=prininit nfactors=10 rotate=hk msa corr priors=smc hey outstat=hk02
plots=(scree initloadings preloadings loadings);
var lm: rh: ew: ha: ia: hc: sc: se;;
run;

*ANALYSIS OF THE RESULTS(QUART02) - EXTRACTION OF THE FACTORS, CROSSLOADS, ETC;
* VARLIST - VARIABLES WITH HIGHEST LOADING OF 0.5;
* CROSSLOADS - TWO OR MORE LOADINGS > 0.3;
* COMMUNALITIES CONTAIN COMMUNALITY TABLE;
data communalities;
    set quart02;
    if _TYPE_ = "COMMUNAL";
    drop _TYPE_ _NAME_;
run;
proc transpose data=communalities out= trancom;
run;
proc sort data = trancom;
    by coll;
run;
data sasuser.communal;
    set trancom;
run;

data loadings;
    set quart02;
    if _TYPE_ = "PATTERN";
    drop _TYPE_;
run;
proc transpose data=loadings out= tranload;
run;
data tranload;
    set tranload;
    high30=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>0.3 then high30+1;
    end;
run;
data varlist;
    set tranload;
    if high30 > 1 then delete;
    if high30 = 0 then delete;
    maxload=0; maxfactor=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>maxload then do; maxload = h(i); maxfactor=i; end;
    end;
run;

data crossloads;
set tranload;
    if high30 in ( 0 1) then delete;
    max1=largest(1,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10) );
    max2=largest(2,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));
    max3=largest(3,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])=max1 then maxfactor1=i;
        if abs(h[i])=max2 then maxfactor2=i;
        if abs(h[i])=max3 then maxfactor3=i;
    end;
    diff=max1-max2;
run;

proc sort data=varlist;
    by maxfactor maxload;
run;
proc sort data=crossloads ;
    by descending diff;
run;
data sasuser.varlist;

```

```

    set varlist;
run;
data sasuser.crossloads;
    set crossloads;
run;
data sasuser.quart02;
    set quart02;
run;
*SCATTERPLOTS FOR FINAL FACTOR SEPARATION;
data outvarlist1;
    set varlist;
        length colorval shapeval $8;
        shapeval="balloon";
        if maxfactor=1 then colorval="blue";
        if maxfactor=2 then colorval="red";
        if maxfactor=3 then colorval="green";
        if maxfactor=9 then colorval="black";
        keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvarlist2;
    set crossloads;
        length colorval shapeval $8;
        shapeval="balloon";
        if diff<0.2 then delete;
        if maxfactor1=1 then colorval="blue";
        if maxfactor1=2 then colorval="red";
        if maxfactor1=3 then colorval="green";
        if maxfactor1>3 then colorval="black";
        keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvar;
    set outvarlist1 outvarlist2;
run;
proc g3d data=outvar;
scatter factor1*factor2=factor3/size=1 noneedle shape='balloon' color=colorval shape=shapeval rotate=30 60 90
120 150 180 210 240 270 300;
title ' Scatterplot for Final Variable List- Quartimax Method';
run;
*****;
*OTHER FACTOR ANALYSIS;
proc factor data=trtp3 method=prinrit nfactors=10 rotate=hk msa corr priors=smc hey outstat=hk02
plots=(scree initloadings preloadings loadings);
var lm: rh: ew: ha: ia: hc: sc: se:;*r5dcontnew--ir5condhome6;
run;
data hkloadings;
    set hk02;
    if _TYPE_ = "PATTERN";
    drop _TYPE_;
run;
proc transpose data=hkloadings out= tranloadhk;
run;
data tranloadhk;
    set tranloadhk;
    high30=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>0.3 then high30+1;
    end;
run;
data varlisthk;
    set tranloadhk;
    if high30 > 1 then delete;
    if high30 = 0 then delete;
    maxload=0;maxfactor=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>maxload then do; maxload = h(i); maxfactor=i; end;
    end;
run;

data crossloadshk;
set tranloadhk;
    if high30 in ( 0 1) then delete;
    max1=largest(1,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10) );
    max2=largest(2,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));
    max3=largest(3,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));

```

```

array h{*} factor1-factor10;
do i = 1 to dim(h) ;
  if abs(h[i])=max1 then maxfactor1=i;
  if abs(h[i])=max2 then maxfactor2=i;
  if abs(h[i])=max3 then maxfactor3=i;
end;
diff=max1-max2;
run;
proc sort data=varlisthk;
  by maxfactor maxload;
run;
proc sort data=crossloadshk ;
  by descending diff;
run;
data sasuser.varlisthk;
  set varlisthk;
run;
data sasuser.crossloadshk;
  set crossloadshk;
run;
data sasuser.hk02;
  set hk02;
run;
*SCATTERPLOTS FOR FINAL FACTOR SEPARATION;
data outvarlist1;
  set varlisthk;
  length colorval shapeval $8;
  shapeval="balloon";
  if maxfactor=1 then colorval="blue";
  if maxfactor=2 then colorval="red";
  if maxfactor=3 then colorval="green";
  if maxfactor=9 then colorval="black";
  keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvarlist2;
  set crossloadshk;
  length colorval shapeval $8;
  shapeval="balloon";
  if diff<0.2 then delete;
  if maxfactor1=1 then colorval="blue";
  if maxfactor1=2 then colorval="red";
  if maxfactor1=3 then colorval="green";
  if maxfactor1>3 then colorval="black";
  keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvar;
  set outvarlist1 outvarlist2;
run;
proc g3d data=outvar;
scatter factor1*factor2=factor3/size=1 noneedle shape='balloon' color=colorval shape=shapeval rotate=30 60 90
120 150 180 210 240 270 300;
title ' Scatterplot for Final Variable List - Oblique Rotation Method';
run;

*PRINCIPAL COMPONENT Analysis;
ods graphics on;
proc princomp data=trtp3 out=ProPC plots=all ;
var lm: rh: ew: ha: ia: hc: sc: se;;
run;

goptions reset=all;
symbol font=, value=dot;
proc gplot data=ProPC;
plot prin2*prin1/ vaxis=axis2 haxis=axis1 nolegend;
axis1 label=("z1" justify=center);
axis2 label=("z2" justify=center r=0 a=90);
run;quit;
*****Factor PRINCIPAL COMPONENT Iterative
*****;
proc factor data=trtp3 method=prin1t nfactors=10 msa corr priors=smc hey outstat=prinit02
plots=(scree initloadings preloadings loadings);
var lm: rh: ew: ha: ia: hc: sc: se;;*r5dcontnew--ir5condhome6;
run;
*ANALYSIS OF THE RESULTS(PRINIT02) - EXTRACTION OF THE FACTORS, CROSSLOADS, ETC;
* VARLIST - VARIABLES WITH HIGHEST LOADING OF 0.5 ;
* CROSSLOADS - TWO OR MORE LOADINGS > 0.3;
* COMMUNALITIES CONTAIN COMMUNALITY TABLE;
data princom;
  set prinit02;

```

```

        if _TYPE_ = "COMMUNAL";
        drop _TYPE_ _NAME_;
run;
proc transpose data=princom out= tranprincom;
run;
proc sort data = tranprincom;
    by coll;
run;
data sasuser.princommunal;
    set tranprincom;
run;

data prinloadings;
    set prinit02;
    if _TYPE_ = "PATTERN";
    drop _TYPE_;
run;
proc transpose data=prinloadings out= tranprinload;
run;
data tranprinload;
    set tranprinload;
    high30=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>0.3 then high30+1;
    end;
run;
data varprinlist;
    set tranprinload;
    if high30 > 1 then delete;
    if high30 = 0 then delete;
    maxload=0;maxfactor=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>maxload then do; maxload = h(i); maxfactor=i; end;
    end;
run;

data crossprinloads;
set tranprinload;
    if high30 in ( 0 1) then delete;
    max1=largest(1,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10) );
    max2=largest(2,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));
    max3=largest(3,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])=max1 then maxfactor1=i;
        if abs(h[i])=max2 then maxfactor2=i;
        if abs(h[i])=max3 then maxfactor3=i;
    end;
    diff=max1-max2;
run;

proc sort data=varprinlist;
    by maxfactor maxload;
run;
proc sort data=crossprinloads ;
    by descending diff;
run;
data sasuser.varprinlist;
    set varprinlist;
run;
data sasuser.crossprinloads;
    set crossprinloads;
run;
data sasuser.prinit02;
    set prinit02;
run;
*SCATTERPLOTS FOR FINAL FACTOR SEPARATION;
data outvarlist1;
    set varprinlist;
    length colorval shapeval $8;
    shapeval="balloon";
    if maxfactor=1 then colorval="blue";
    if maxfactor=2 then colorval="red";
    if maxfactor=3 then colorval="green";

```



```

        if maxfactor=9 then colorval="black";
        keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvarlist2;
    set crossprinloads;
    length colorval shapeval $8;
    shapeval="balloon";
    if diff<0.2 then delete;
    if maxfactor1=1 then colorval="blue";
    if maxfactor1=2 then colorval="red";
    if maxfactor1=3 then colorval="green";
    if maxfactor1>3 then colorval="black";
    keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;

run;
data outvar;
    set outvarlist1 outvarlist2;
run;
proc g3d data=outvar;
scatter factor1*factor2=factor3/size=1 noneedle shape='balloon' color=colorval shape=shapeval rotate=30 60 90
120 150 180 210 240 270 300;
title ' Scatterplot for Final Variable List - Principal Iterative Method';
run;

*****ML METHOD *****;
proc factor data=trtp3 method=ml nfactors=10 msa corr priors=smc hey outstat=ml02
plots=(scree initloadings preloadings loadings);
var lm: rh: ew: ha: ia: hc: sc: se:;*r5dcontnew--ir5condhome6;
run;
*ANALYSIS OF THE RESULTS(PRINIT02) - EXTRACTION OF THE FACTORS, CROSSLOADS, ETC;
* VARLIST - VARIABLES WITH HIGHEST LOADING OF 0.5 ;
* CROSSLOADS - TWO OR MORE LOADINGS > 0.3;
* COMMUNALITIES CONTAIN COMMUNALITY TABLE;
data mlcom;
    set ml02;
    if _TYPE_ = "COMMUNAL";
    drop _TYPE_ _NAME_;
run;
proc transpose data=mlcom out= tranmlcom;
run;
proc sort data = tranmlcom;
    by coll;
run;
data sasuser.mlcommunal;
    set tranmlcom;
run;

data mlloadings;
    set ml02;
    if _TYPE_ = "PATTERN";
    drop _TYPE_;
run;
proc transpose data=mlloadings out= tranmlload;
run;
data tranmlload;
    set tranmlload;
    high30=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>0.3 then high30+1;
    end;
run;
data varmllist;
    set tranmlload;
    if high30 > 1 then delete;
    if high30 = 0 then delete;
    maxload=0;maxfactor=0;
array h{*} factor1-factor10;
    do i = 1 to dim(h) ;
        if abs(h[i])>maxload then do; maxload = h(i); maxfactor=i; end;
    end;
run;

data crossmlloads;
set tranmlload;
    if high30 in ( 0 1) then delete;
    max1=largest(1,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10) );
    max2=largest(2,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
        abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));

```

```

max3=largest(3,abs(factor1),abs(factor2), abs(factor3), abs(factor4), abs(factor5),
abs(factor6), abs(factor7), abs(factor8), abs(factor9), abs(factor10));
array h{*} factor1-factor10;
do i = 1 to dim(h) ;
  if abs(h[i])=max1 then maxfactor1=i;
  if abs(h[i])=max2 then maxfactor2=i;
  if abs(h[i])=max3 then maxfactor3=i;
end;
diff=max1-max2;
run;

proc sort data=varmlist;
  by maxfactor maxload;
run;
proc sort data=crossmlloads ;
  by descending diff;
run;
data sasuser.varmlist;
  set varmlist;
run;
data sasuser.crossmlloads;
  set crossmlloads;
run;
data sasuser.ml02;
  set ml02;
run;
*SCATTERPLOTS FOR FINAL FACTOR SEPARATION;
data outvarlist1;
  set varmlist;
  length colorval shapeval $8;
  shapeval="balloon";
  if maxfactor=1 then colorval="blue";
  if maxfactor=2 then colorval="red";
  if maxfactor=3 then colorval="green";
  if maxfactor=9 then colorval="black";
  keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvarlist2;
  set crossmlloads;
  length colorval shapeval $8;
  shapeval="balloon";
  if diff<0.2 then delete;
  if maxfactor1=1 then colorval="blue";
  if maxfactor1=2 then colorval="red";
  if maxfactor1=3 then colorval="green";
  if maxfactor1>3 then colorval="black";
  keep _NAME_ _LABEL_ factor1 factor2 factor3 colorval shapeval;
run;
data outvar;
  set outvarlist1 outvarlist2;
run;
proc g3d data=outvar;
scatter factor1*factor2=factor3/size=1 noneedle shape='balloon' color=colorval shape=shapeval rotate=30 60 90
120 150 180 210 240 270 300;
title ' Scatterplot for Final Variable List - ML Method';
run;
***** THE END *****;

```